Figures

Overview

Since this is a relatively small project with relatively simple figures, I'll try just making all the figures in one notebook from hard-coded numbers, and we'll see how that goes.

Bar chart

```
library(ggplot2)

source("utils.R")

# Read config.yml
cfg <- config::get()
```

```
df <- data.frame(
  category = c(
    "Tech Transfer",
    "Library",
    "Administration",
    "An academic dept",
    "An RSE group",
    "A research center",
    "An AI institute",
    "An HPC center"
  ),
  value = c(8, 7, 5, 4, 3, 3, 3, 2)
)
df <- df[order(-df$value), ]
df$category <- factor(df$category, levels = unique(df$category))
```

```r
# Custom hex palette
# custom_colors <- c(
#   "#00817f",
#   "#42F2FF",
#   "#B92DA4",
#   "#F7D00B",
#   "#124B5C",
#   "#6A1B70",
#   "#F27405",
#   "#D9A00E"
# )


ggplot(df, aes(x = category, y = value)) +
  geom_col(width = 0.6, fill = "#39a695") +
  scale_fill_manual(values = "#39a695") +
  # Anchor bars flush to the x-axis with 10% headroom at the top
  scale_y_continuous(expand = expansion(mult = c(0, 0.1))) +
  labs(
    title = "OSPO locations",
    x = NULL,
    y = "Number of OSPOs"
  ) +

theme(
    # Canvas backgrounds
    plot.background = element_rect(fill = "#FFFFFF", color = NA),
    panel.background = element_rect(fill = "#FFFFFF", color = NA),

    # Subtle horizontal gridlines only
    panel.grid.major.y = element_line(color = "#E5E7EB", linewidth = 0.5),
    panel.grid.major.x = element_blank(),
    panel.grid.minor = element_blank(),

    # Clean baseline axis without tick marks
    axis.line.x = element_line(color = "#374151", linewidth = 0.6),
    axis.line.y = element_blank(),
    axis.ticks = element_blank(),

    # Typography and spacing
    plot.margin = margin(t = 20, r = 20, b = 15, l = 15),
    plot.title = element_text(color = "#111827", size = 14, face = "bold", margin = margin(b
```

```
    axis.title.y = element_text(color = "#374151", size = 11, margin = margin(r = 10)),
    axis.text.x = element_text(color = "#374151", size = 10, angle = 45, hjust = 1, vjust = 1

    # Remove redundant legend
    legend.position = "none"
  )
```

```
ggsave(
  filename = "bar_chart.png",
  width = 4.5,
  height = 3.2,
  units = "in",
  dpi = 300,
  bg = "#FFFFFF"
)
```

Donut chart

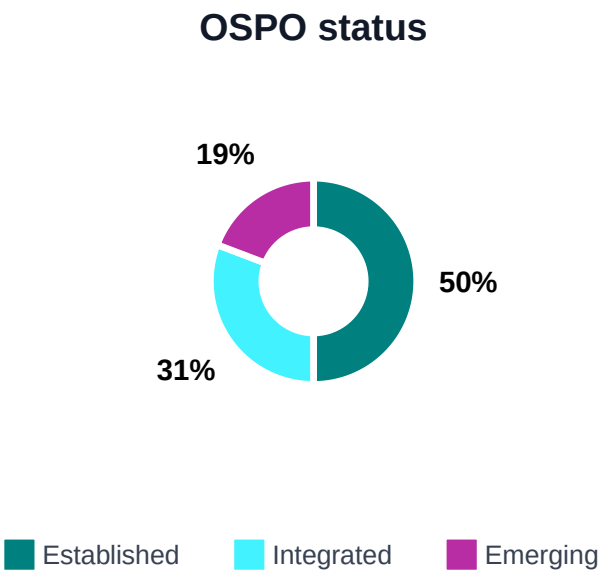
```
df <- data.frame(
  category = c("Established", "Integrated", "Emerging"),
```

```r
  value = c(13, 8, 5)
)

# Preserve category order and compute slice coordinates
df$category <- factor(df$category, levels = df$category)
df$fraction <- df$value / sum(df$value)
df$ymax <- cumsum(df$fraction)
df$ymin <- c(0, head(df$ymax, n = -1))
df$label_pos <- (df$ymax + df$ymin) / 2
df$label <- paste0(round(df$fraction * 100), "%")

# Custom hex palette for the three sections
custom_colors <- c(
  "#00817f",
  "#42F2FF",
  "#B92DA4"
)
ggplot(df, aes(ymax = ymax, ymin = ymin, xmax = 4, xmin = 3, fill = category)) +
  geom_rect(color = "#FFFFFF", linewidth = 1.2) +
  # Move labels further past the outer edge (xmax = 4) for clear separation
  geom_text(
    aes(x = 5, y = label_pos, label = label),
    color = "#000000",
    fontface = "bold",
    size = 4
  ) +
  coord_polar(theta = "y", clip = "off") +
  # Expand upper radial limit to match the wider label offset
  xlim(c(2, 5.2)) +
  scale_fill_manual(values = custom_colors) +
  labs(
    title = "OSPO status",
    fill = NULL
  ) +
  # Custom theme built directly via theme() without pre-made presets
  theme(
    plot.background = element_rect(fill = "#FFFFFF", color = NA),
    panel.background = element_rect(fill = "#FFFFFF", color = NA),
    panel.grid = element_blank(),
    axis.title = element_blank(),
    axis.text = element_blank(),
    axis.ticks = element_blank(),
```

```
    axis.line = element_blank(),
    plot.margin = margin(t = 20, r = 20, b = 15, l = 20),
    plot.title = element_text(
      color = "#111827",
      size = 14,
      face = "bold",
      hjust = 0.5,
      margin = margin(b = 10)
    ),
    legend.position = "bottom",
    legend.background = element_rect(fill = "#FFFFFF", color = NA),
    legend.key = element_rect(fill = "#FFFFFF", color = NA),
    legend.text = element_text(color = "#374151", size = 10, margin = margin(r = 12))
  )
```




```
ggsave(
  filename = "donut_chart.png",
  width = 4,
  height = 4,
  units = "in",
  dpi = 300,
  bg = "#FFFFFF"
```

```
)
```